

PHETCHABUN, Thailand

WMO: 483790

Lat: 16.4347N

Lon: 101.1517E

Elev: 381

StdP: 14.49

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	58.9	61.6	48.6	51.5	67.8	52.4	59.2	70.8	12.1	76.4	10.7	77.2	0.3	10	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.6	101.3	79.1	99.2	78.7	97.2	78.4	82.5	93.7	81.7	92.3	81.0	91.1	3.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.9	157.2	88.4	79.1	153.1	86.9	78.6	150.4	86.1	46.9	93.8	45.9	92.8	45.2	91.6	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.4	7.3	DB	53.8	104.2	4.0	1.8	50.9	105.5	48.6	106.6	46.3	107.6	43.4	108.9	(4)
(5)			WB	51.1	84.5	3.8	1.9	48.3	85.8	46.1	86.9	44.0	88.0	41.2	89.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	77.4	81.1	85.0	87.4	85.6	84.7	83.0	82.5	82.6	82.3	80.2	77.0	(6)	
	DBStd	4.36	3.86	3.81	3.73	3.49	3.16	2.48	2.33	2.00	1.95	2.24	3.35	4.27	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD50	11825	850	870	1085	1121	1105	1040	1024	1008	978	1002	906	836	(10)	
	CDD65	6351	385	450	620	671	640	590	559	543	528	537	456	372	(11)	
	CDH74	69208	3666	4876	7571	8628	7566	6888	6022	5556	5232	5348	4419	3436	(12)	
	CDH80	31545	1628	2533	4219	4875	3641	3074	2335	1947	1805	2126	1927	1435	(13)	
(14) Wind	WSAvg	1.5	1.2	1.2	1.6	1.6	1.5	1.7	1.7	1.4	1.0	1.3	1.6	1.6	(14)	
(15) Precipitation	PrecAvg	45.2	0.4	0.6	1.5	2.9	6.2	5.3	7.1	8.3	8.4	4.0	0.6	0.1	(15)	
	PrecMax	69.8	2.6	1.7	4.6	7.3	11.2	11.0	11.5	13.5	14.4	10.9	2.5	0.7	(16)	
	PrecMin	31.3	0.0	0.0	0.0	0.0	2.4	2.4	3.9	2.2	4.2	0.7	0.0	0.0	(17)	
	PrecStd	10.2	0.8	0.6	1.3	1.9	2.4	2.6	2.3	2.7	3.4	2.7	0.6	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.1	99.0	102.5	105.2	102.0	98.6	96.1	93.6	93.5	94.6	95.3	94.6	(19)	
		MCWB	73.9	76.1	77.3	80.3	81.0	79.8	79.7	80.8	80.2	78.5	77.8	75.6	(20)	
	2%	DB	92.9	96.7	100.3	102.4	98.6	95.4	92.9	91.5	91.6	92.7	93.4	92.1	(21)	
		MCWB	73.6	75.6	77.3	79.7	80.9	80.2	80.2	80.5	79.9	78.4	76.9	74.5	(22)	
	5%	DB	91.0	94.9	98.3	99.9	96.0	93.2	90.9	89.9	90.0	91.3	91.6	90.1	(23)	
		MCWB	73.1	75.3	77.1	79.7	80.7	80.2	79.9	80.0	79.7	77.8	76.2	73.5	(24)	
	10%	DB	88.8	92.7	96.2	97.4	93.3	91.2	89.2	88.2	88.2	89.6	89.5	87.9	(25)	
		MCWB	72.1	74.8	76.9	79.4	80.4	80.0	79.6	79.6	79.3	77.5	75.0	72.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	80.0	81.2	83.8	84.1	82.8	82.5	82.2	82.1	81.2	80.4	78.4	(27)	
		MCDB	90.1	94.5	96.8	99.2	96.6	91.7	90.6	90.6	90.5	90.4	91.4	89.8	(28)	
	2%	WB	76.5	78.5	79.8	81.9	82.5	81.7	81.3	81.1	81.1	80.2	79.0	76.7	(29)	
		MCDB	88.1	93.0	95.8	97.5	94.1	91.0	89.4	89.0	88.9	88.8	89.4	87.9	(30)	
	5%	WB	75.1	77.4	78.9	80.9	81.5	81.0	80.6	80.5	80.4	79.5	77.9	75.4	(31)	
		MCDB	86.1	91.3	94.7	96.2	92.6	90.2	88.8	88.1	87.8	87.7	87.5	86.0	(32)	
	10%	WB	73.7	76.1	77.9	79.9	80.7	80.4	79.9	79.9	79.8	78.7	76.7	74.0	(33)	
		MCDB	84.4	89.0	92.9	94.5	91.3	89.3	87.5	86.8	86.5	86.4	85.5	84.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	24.2	24.7	23.3	20.6	16.1	14.2	12.7	12.2	12.7	15.5	19.3	22.2	(35)	
		MCDBR	26.5	26.9	26.4	23.5	19.7	16.7	15.2	14.4	14.9	17.8	21.0	23.9	(36)	
	5% WB	MCWBR	9.8	9.7	9.0	7.5	6.2	5.3	5.4	5.4	5.4	5.7	6.9	8.5	(37)	
		MCWBR	22.0	24.0	23.5	21.5	17.5	14.9	13.7	13.4	13.7	15.4	18.8	21.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.486	0.591	0.671	0.619	0.505	0.453	0.442	0.451	0.491	0.538	0.464	0.465	(40)	
		taud	2.073	1.829	1.683	1.837	2.182	2.349	2.377	2.336	2.209	2.036	2.228	2.160	(41)	
	Ebn at Noon		253	232	219	231	255	266	270	270	260	243	259	256	(42)	
		Edh at Noon	50	67	80	68	48	40	39	41	46	54	43	45	(43)	
(44) All-Sky Solar Radiation	RadAvg		1498	1628	1679	1808	1699	1597	1484	1432	1424	1487	1534	1470	(44)	
	RadStd		102	75	104	87	100	111	86	76	84	106	70	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	+376	+382
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	+0.25	+0.29	N/A	N/A	+191	+193

Nomenclature: See separate page